

# Hisu Kim

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## EDUCATION

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| <b>Gwangju Institute of Science and Technology, South Korea</b><br><i>M.S. in Climate Analysis and Modeling Lab (Advisor: Jin-ho Yoon)</i>                                                                 | Sep. 2025 – Aug. 2027 (expected)<br>TGPA 4.37/4.5 |
| <b>Gwangju Institute of Science and Technology, South Korea</b><br><i>B.S. in Electrical Engineering and Computer Science<br/>and Earth Science and Environmental Engineering<br/>Minor in Mathematics</i> | 2019 – 2025<br>TGPA 4.10/4.5                      |
| <b>Warsaw University of Technology, Poland</b><br><i>Exchange student, Faculty of Building Services, Hydro and Environmental Engineering</i>                                                               | Oct. 2022 – Feb. 2023<br>TGPA 4.86/5.0            |
| <b>University of California, Berkeley, USA</b><br><i>Summer Session C</i>                                                                                                                                  | June – Aug. 2023<br>TGPA 4.0/4.0                  |

## PUBLICATIONS AND CONFERENCES

### Publications

- **Kim, H.**, Ryu, J., Son, S., Jeong, J., Kim, H., & Yoon, J.-H. (2026). A spectral test of the butterfly effect and physical consistency in the diffusion-based GenCast's ensembles. *npj Climate and Atmospheric Science*. <https://doi.org/10.1038/s41612-026-01380-1>
- Ryu, J., **Kim, H.**, Wang, S.-Y., & Yoon, J.-H. (2026). Increasing resolution and accuracy in sub-seasonal forecasting through 3D U-Net: The western US. *Geoscientific Model Development* 19(1), 27–39. <https://doi.org/10.5194/gmd-19-27-2026>

### Conferences

- **Kim, H.**, Ryu, J., Jang, W., Kwon, M., & Yoon, J. (2026, September). Uncertainty Assessment in Ensemble Forecast Post-processing Using U-Net and Diffusion. *Accepted at S2S2D 2026*, Reading, United Kingdom
- **Kim, H.**, & Yoon, J.-H. (2025, December 16). Revealing the butterfly effect in deep learning weather prediction model via kinetic energy spectra change. *Oral presentation at AGU25 Meeting*, New Orleans, LA, United States.

## PROFESSIONAL DEVELOPMENT

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Rosbypalooza 2026</b> <ul style="list-style-type: none"><li>• University of Chicago, IL, USA</li></ul>                                                                            | 20 – 31 July 2026 |
| <b>Advanced School on Foundation Models for Scientific Discovery</b> <ul style="list-style-type: none"><li>• International Centre for Theoretical Physics, Italy (virtual)</li></ul> | 10 – 18 July 2025 |
| <b>Summer School of Mathematics: Algorithms</b> <ul style="list-style-type: none"><li>• Eötvös Loránd University, Hungary</li></ul>                                                  | 23 – 27 June 2025 |

## SCHOLARSHIPS AND AWARDS

|                                                                                                                                                                                                    |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>2026 Weather Big Data Contest</b>   <i>Korea Meteorological Administration</i> <ul style="list-style-type: none"><li>• Grand Prize, Minister of Climate, Energy and Environment Award</li></ul> | Aug. 2026             |
| <b>Presidential Science Scholarship for Graduate Student</b>   <i>Ministry of Science and ICT</i>                                                                                                  | 2026-7 (1-year)       |
| <b>Government Funded Scholarship for GIST Student</b>   <i>GIST</i>                                                                                                                                | 2019 Spring – Present |
| <b>Scholarship for Outbound Program</b>   <i>GIST</i>                                                                                                                                              | 2023 Fall             |
| <b>Academic Excellence Scholarship</b>   <i>GIST</i>                                                                                                                                               | 4 Semesters           |
| <b>Scholarship for Summer Session Program Abroad</b>   <i>GIST</i>                                                                                                                                 | 2023 Summer           |

TEACHING EXPERIENCE

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- Calculus (Single- & Multivariable) | *Teaching Assistant*

2025 Spring – 2026 Fall

  - Single-variable (2026 Spring): designed weekly problem sets, prepared and led recitation sessions, and provided written feedback on students’ solutions
  - Multivariable (2025 Spring & Fall, 2026 Fall): held weekly office hours, answering conceptual questions and guiding students through theorem proofs
- Graph Theory | *Teaching Assistant*

2024 Fall

  - Conducted weekly office hours, answering questions and guiding students through theorem proofs
  - Organized regular sessions to provide detailed feedback on homework and quizzes, and advanced mathematical  $\text{\LaTeX}$  usage
  - Offered supplementary lessons on challenging topics and extra problems
- Literature Courses from Prof. Soo-Jeong Lee | *Teaching Assistant*

2023 Spring – 2026 Spring

  - Built syllabus generator website to streamline the professor’s course preparation process
  - Used  $\text{\LaTeX}$  to edit e-books compiled from students’ story-writing assignments
- Digital Design | *Teaching Assistant*

2023 Spring

  - Conducted office hours twice a week answering questions from homework
  - Created sample exam questions to facilitate student learning

WORK EXPERIENCE

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- Cybersecurity Analyst

Feb. 2021 – Oct. 2022

*Republic of Korea Naval Fleet Command*

*Computer Emergency Response Team*

  - Co-administered dozens of Linux production servers across the Fleet Command network
  - Mitigated *log4j* and *MS-MSDT* vulnerabilities via NAC-distributed registry patches and removal of exploitable files; analyzed firewall/IPS packets and logs to flag unauthorized or anomalous communications

SKILLS

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|             |                                              |                |
|-------------|----------------------------------------------|----------------|
| Programming | Python                                       | (Main)         |
|             | C/C++, C#, JavaScript, Bash                  | (Intermediate) |
| Libraries   | PyTorch (Lightning), JAX, xarray, Dask, zarr |                |
| Tools       | Linux, Git, $\text{\LaTeX}$ , NCL, GrADS     |                |
| Language    | Korean (native)                              |                |
|             | English (fluent)                             |                |
|             | Spanish, Polish (elementary)                 |                |